

# QC Division API Contract

Case Prep / RMP partial-approval style

## Aligned with

- Endpoints the UI already calls
- Frontend create payload: `divisionDetails[]`.`data.sections[]` (`mapQualityControlPayload`) with `motorId` / `premixNo` / `subType` on sections
- Case Prep pattern: unit DRAFT / SUBMIT while root `formSubmissionType` stays DRAFT until final proceed
- Extra QC layer: unit -> division -> batch

## Endpoints (current UI)

| Action  | Method | Path                             |
|---------|--------|----------------------------------|
| Create  | POST   | /api/v1/user/qc-division/create  |
| Update  | PUT    | /api/v1/user/qc-division/update  |
| Details | POST   | /api/v1/user/qc-division/details |

## Enums

| Field                                                                                                  | Values                                                                                                                        |
|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <code>formSubmissionType</code>                                                                        | DRAFT, SUBMIT                                                                                                                 |
| <code>motorSubmissionType</code>                                                                       | DRAFT, SUBMIT                                                                                                                 |
| <code>premixSubmissionType</code>                                                                      | DRAFT, SUBMIT                                                                                                                 |
| <code>divisionSubmissionType</code>                                                                    | DRAFT, SUBMIT (only for division-scoped flows, e.g. revalidation)                                                             |
| <code>motorSubmissionStatus</code> / <code>premixSubmissionStatus</code> / <code>divisionStatus</code> | TO_BE_INITIATED, IN_PROGRESS, WAITING_FOR_APPROVAL, APPROVED, REJECTED                                                        |
| Form/batch status                                                                                      | DRAFT, IN_PROGRESS, WAITING_FOR_PARTIAL_APPROVAL, WAITING_FOR_COMPLETE_APPROVAL, APPROVED, REJECTED, FINAL_APPROVAL_COMPLETED |

Do not use FINAL\_SUBMIT / SUBMITTED - match RMP/Case Prep (DRAFT / SUBMIT).

## Approval hierarchy

| Level                                  | Approved when                                                                                           |
|----------------------------------------|---------------------------------------------------------------------------------------------------------|
| Unit (motor / premix / final-mix card) | Approver accepts that unit                                                                              |
| Division                               | Every unit in that division is APPROVED                                                                 |
| Batch (QC form)                        | Every division on the form is APPROVED, then user does final <code>formSubmissionType</code> : "SUBMIT" |

- `formSubmissionType` stays DRAFT for all unit draft/submit calls.
- Final proceed uses `formSubmissionType`: "SUBMIT" only when `allDivisionsApproved === true`.

## Payload model (important)

Frontend today builds:

```
divisionDetails: [{
  division,           // QcApiDivision
  subType,            // QcApiSubType | null
  data: { sections } // SchemaSectionSubmission[] (+ motorId | premixNo | subType on each section)
}]
```

SchemaSectionSubmission:

```
{ sectionId: string; sectionData: unknown[]; motorId?: string; premixNo?: number; subType?: string; ... }
```

Recommended wire format (Case Prep mirror): wrap units explicitly under data, keep sections identical to schema-engine output.

```
"data": {
  "motors": [
    {
      "motorId": "MOTOR-01",
      "motorSubmissionType": "DRAFT",
      "sections": [
        { "sectionId": "CASTING_SELECTION", "sectionData": [{ "...": "..."} ] }
      ]
    }
  ]
}
```

- Premix divisions use `premixes[]` + `premixSubmissionType`.
- Backend may still accept the current flat `data.sections[]` (with `motorId/premixNo` on sections) during migration; details must return the wrapped shape so partial nav can hydrate cleanly.
- Send only the division + unit being acted on.

## Division -> unit matrix

| division                  | subType (examples)                                | Unit key           | Submission field       |
|---------------------------|---------------------------------------------------|--------------------|------------------------|
| RAW_MATERIAL_REVALIDATION | null                                              | Division itself    | divisionSubmissionType |
| RAW_MATERIAL_PROCESSING   | SOLID_PROCESSING / LIQUID_PROCESSING / null       | premixNo           | premixSubmissionType   |
| MIXING                    | PREMIX / FINAL_MIX                                | premixNo (+ stage) | premixSubmissionType   |
| HARDWARE                  | ABRADING / PREHEATING / LINEAR_COATING / DISPATCH | motorId            | motorSubmissionType    |
| CASTING                   | null                                              | motorId            | motorSubmissionType    |
| CURING                    | NORMAL / CONFINED / N2_PRESSURE                   | motorId            | motorSubmissionType    |
| DE_CORING                 | null                                              | motorId            | motorSubmissionType    |
| TRIMMING                  | MAIN_BATCH / SUBSCALE                             | motorId            | motorSubmissionType    |
| POST_CURE                 | LOOSE_FLAP_FILLING / INHIBITION                   | motorId            | motorSubmissionType    |
| NDT                       | null                                              | motorId            | motorSubmissionType    |
| PROPELLANT_PROPERTIES     | MECHANICAL_PROPERTIES / ...                       | motorId            | motorSubmissionType    |
| WEIGHTMENT                | null                                              | motorId            | motorSubmissionType    |

## 1. Create Form

POST /api/v1/user/qc-division/create

### Motor division - Save Motor Draft / Submit Motor

- `formSubmissionType` stays DRAFT until final proceed.
- Send only the motor being acted on.

#### Create - CASTING motor DRAFT

```
{
  "batchId": "BATCH-2026-001",
  "subDepartmentId": 115,
  "formSubmissionType": "DRAFT",
  "divisionDetails": [
    {
      "division": "CASTING",
      "subType": null,
      "data": {
```

```

"motors": [
  {
    "motorId": "MOTOR-01",
    "motorSubmissionType": "DRAFT",
    "sections": [
      {
        "sectionId": "CASTING_SELECTION",
        "sectionData": [
          { "CASTING_TYPE": "SINGLE" }
        ]
      },
      {
        "sectionId": "FINAL_ASSEMBLY",
        "sectionData": [
          {
            "ASSEMBLY_DATE": "2026-06-15",
            "MANDREL_ASSEMBLY": {
              "rows": [
                {
                  "SR_NO": 1,
                  "READING_WITHOUT_CUP": "10.5",
                  "READING_WITH_BOTTOM_CUP": "12.2"
                }
              ]
            }
          }
        ]
      },
      {
        "sectionId": "PROPELLANT_CASTING",
        "sectionData": [{ "...": "schema fields / tables" }]
      },
      {
        "sectionId": "WEIGHTMENT_DETAILS",
        "sectionData": [{ "...": "..." }]
      },
      {
        "sectionId": "POST_CAST_OPERATIONS",
        "sectionData": [{ "...": "..." }]
      }
    ]
  }
]
}

```

Submit Motor - same body, only change:

```
"motorSubmissionType": "SUBMIT"
```

## Create - MIXING premix SUBMIT

```

{
  "batchId": "BATCH-2026-001",
  "subDepartmentId": 115,
  "formSubmissionType": "DRAFT",
  "divisionDetails": [
    {
      "division": "MIXING",
      "subType": "PREMIX",
      "data": {
        "premises": [
          {
            "premixNo": 1,
            "premixSubmissionType": "SUBMIT",
            "sections": [
              {
                "sectionId": "PREMIX_DETAILS",
                "sectionData": [
                  {
                    "PREMIX_DETAILS": {
                      "rows": [
                        {
                          "PARAMETER": "Homogeneity",
                          "BOWL_NO": "PM-001",
                          "DATE_OF_PREMIX": "2026-06-15",

```

Create - RAW\_MATERIAL\_PROCESSING premix DRAFT

```
{
  "batchId": "BATCH-2026-001",
  "subDepartmentId": 115,
  "formSubmissionType": "DRAFT",
  "divisionDetails": [
    {
      "division": "RAW_MATERIAL_PROCESSING",
      "subType": "SOLID_PROCESSING",
      "data": {
        "premises": [
          {
            "premixNo": 1,
            "premixSubmissionType": "DRAFT",
            "sections": [
              {
                "sectionId": "SOLID_PROCESSING_DETAILS",
                "sectionData": [{ "...": "solid schema sections" }],
                "subType": "SOLID_PROCESSING"
              }
            ]
          }
        ]
      }
    }
  ]
}
```

For BOTH solid+liquid on same premix, either:

1. one create/update with two section groups (subType per section), or
2. two calls with subType: "SOLID\_PROCESSING" / "LIQUID\_PROCESSING" (UI already builds both section sets for BOTH\_PREMIX).

Create - RAW\_MATERIAL\_REVALIDATION (division-scoped, no motors/premixes)

```
{
  "batchId": "BATCH-2026-001",
  "subDepartmentId": 115,
  "formSubmissionType": "DRAFT",
  "divisionDetails": [
    {
      "division": "RAW_MATERIAL_REVALIDATION",
      "subType": null,
      "divisionSubmissionType": "SUBMIT",
      "data": {
        "sections": [
          {
            "sectionId": "RAW_MATERIAL_DETAILS",
            "sectionData": [
              {
                "RAW_MATERIAL_DETAILS": {
                  "rows": [
                    {
                      "INGREDIENT": "HTPB",
                      "LOT_BATCH_NUMBER": "LOT-HTPB-001",
                      "PARAMETER": "OH Value, mg KOH/g",
                      "SPECIFICATION": "40-50",
                      "RESULT": "45"
                    }
                  ]
                }
              }
            ]
          }
        ]
      }
    }
  ]
}
```

**Proceed for Final Approval (all divisions already APPROVED)**

```
{
  "batchId": "BATCH-2026-001",
  "subDepartmentId": 115,
  "formSubmissionType": "SUBMIT",
  "divisionDetails": []
}
```

(Or resend full saved divisionDetails from details - same as RMP/Case Prep final submit.)

### Create - Success response

```
{
  "success": true,
  "statusCode": 201,
  "message": "QC division form saved successfully",
  "timestamp": "2026-06-01T20:00:00Z",
  "data": {
    "formId": "QC-F-2026-0001",
    "batchId": "BATCH-2026-001",
    "subDepartmentId": 115,
    "formSubmissionType": "DRAFT",
    "status": "IN_PROGRESS",
    "division": "CASTING",
    "subType": null,
    "motorId": "MOTOR-01",
    "motorSubmissionType": "DRAFT",
    "motorSubmissionStatus": "IN_PROGRESS",
    "divisionStatuses": [
      {
        "division": "CASTING",
        "subType": null,
        "status": "IN_PROGRESS",
        "allUnitsApproved": false,
        "pendingUnitCount": 0,
        "approvedUnitCount": 0,
        "rejectedUnitCount": 0,
        "waitingForApprovalUnitCount": 0,
        "inProgressUnitCount": 1,
        "totalUnitCount": 2
      }
    ],
    "motorStatuses": [
      {
        "division": "CASTING",
        "motorId": "MOTOR-01",
        "motorSubmissionType": "DRAFT",
        "motorSubmissionStatus": "IN_PROGRESS"
      },
      {
        "division": "CASTING",
        "motorId": "MOTOR-02",
        "motorSubmissionType": null,
        "motorSubmissionStatus": "TO_BE_INITIATED"
      }
    ],
    "premixStatuses": [],
    "allDivisionsApproved": false,
    "pendingDivisionCount": 1,
    "approvedDivisionCount": 0,
    "totalDivisionCount": 1,
    "allMotorsApproved": false,
    "pendingMotorCount": 0,
    "approvedMotorCount": 0,
  }
}
```

```

    "rejectedMotorCount": 0,
    "totalMotorCount": 2,
    "submittedBy": "EMP001",
    "submittedAt": "2026-06-01T20:00:00Z"
  }
}

```

After motor SUBMIT ->

- status: "WAITING\_FOR\_PARTIAL\_APPROVAL"
- motorSubmissionStatus: "WAITING\_FOR\_APPROVAL"
- division status: "WAITING\_FOR\_PARTIAL\_APPROVAL"

For premix calls, echo premixNo / premixSubmissionType / premixStatuses instead of motor fields.

## Create - Errors

Keep existing 400 / 401 / 403 / 404 batch / 404 schema / 422 / 500. Soften 409:

```

{
  "success": false,
  "statusCode": 409,
  "message": "QC division form already exists",
  "error": {
    "code": "QC_FORM_ALREADY_EXISTS",
    "details": "Use update API. Form QC-F-2026-0001 already exists for this batch"
  },
  "data": null
}

```

## 2. Update Form

PUT /api/v1/user/qc-division/update

Same divisionDetails shape as create. Identify form by formId.

```

{
  "formId": "QC-F-2026-0001",
  "batchId": "BATCH-2026-001",
  "subDepartmentId": 115,
  "formSubmissionType": "DRAFT",
  "divisionDetails": [
    {
      "division": "CASTING",
      "subType": null,
      "data": {
        "motors": [
          {
            "motorId": "MOTOR-01",
            "motorSubmissionType": "SUBMIT",
            "sections": [
              { "sectionId": "CASTING_SELECTION", "sectionData": [{ "...": "..." }] },
              { "sectionId": "FINAL_ASSEMBLY", "sectionData": [{ "...": "..." }] }
            ]
          }
        ]
      }
    }
  ]
}

```

### HARDWARE (process subType per motor)

```

{
  "formId": "QC-F-2026-0001",
  "batchId": "BATCH-2026-001",
  "subDepartmentId": 115,
  "formSubmissionType": "DRAFT",
  "divisionDetails": [
    {
      "division": "HARDWARE",
      "subType": "ABRADING",
      "data": {

```

```

    "motors": [
      {
        "motorId": "MOTOR-01",
        "motorSubmissionType": "SUBMIT",
        "sections": [
          {
            "sectionId": "ABRADING_DETAILS",
            "sectionData": [
              {
                "FIRST_CUT": { "rows": [] },
                "SECOND_CUT": { "rows": [] }
              }
            ],
            "subType": "ABRADING"
          }
        ]
      }
    ]
  }
}

```

## CURING / TRIMMING / POST\_CURE / NDT / WEIGHTMENT / PROPELLANT

Same motor wrapper; keep schema sectionIds from that division's schema. Extra metadata the UI already attaches on sections can live on the motor object:

```

{
  "motorId": "MOTOR-01",
  "motorSubmissionType": "DRAFT",
  "motorReceivedDate": "2026-06-28",
  "motorCount": 2,
  "inhibitorType": "IR1",
  "weighscaleNo": "WS-01",
  "calibrationDueDate": "2026-12-31",
  "sections": []
}

```

## Final proceed

```

{
  "formId": "QC-F-2026-0001",
  "batchId": "BATCH-2026-001",
  "subDepartmentId": 115,
  "formSubmissionType": "SUBMIT",
  "divisionDetails": []
}

```

Allowed only if every division is APPROVED (i.e. all units in all divisions approved).

## Update - Rules

1. Upsert by (division, subType?, motorId | premixNo).
2. Locked units (WAITING\_FOR\_APPROVAL / APPROVED) reject section edits -> UNIT\_NOT\_EDITABLE.
3. First create for a batch only; later units use update (QC\_FORM\_ALREADY\_EXISTS if create repeated).
4. formSubmissionType: "SUBMIT" only when allDivisionsApproved === true.
5. Approving all units in a division flips that division to APPROVED automatically.
6. Rejected unit becomes editable again; division/batch leave APPROVED until fixed.

## Update - Success response

```

{
  "success": true,
  "statusCode": 200,
  "message": "QC division form updated successfully",
  "timestamp": "2026-06-01T20:30:00Z",
  "data": {
    "formId": "QC-F-2026-0001",
    "batchId": "BATCH-2026-001",
    "subDepartmentId": 115,
    "formSubmissionType": "DRAFT",
  }
}

```

```

"status": "WAITING_FOR_PARTIAL_APPROVAL",
"division": "CASTING",
"subType": null,
"motorId": "MOTOR-01",
"motorSubmissionType": "SUBMIT",
"motorSubmissionStatus": "WAITING_FOR_APPROVAL",
"divisionStatuses": [
  {
    "division": "CASTING",
    "subType": null,
    "status": "WAITING_FOR_PARTIAL_APPROVAL",
    "allUnitsApproved": false,
    "pendingUnitCount": 1,
    "approvedUnitCount": 0,
    "rejectedUnitCount": 0,
    "waitingForApprovalUnitCount": 1,
    "inProgressUnitCount": 0,
    "totalUnitCount": 2
  }
],
"motorStatuses": [
  {
    "division": "CASTING",
    "motorId": "MOTOR-01",
    "motorSubmissionType": "SUBMIT",
    "motorSubmissionStatus": "WAITING_FOR_APPROVAL"
  },
  {
    "division": "CASTING",
    "motorId": "MOTOR-02",
    "motorSubmissionType": null,
    "motorSubmissionStatus": "TO_BE_INITIATED"
  }
],
"premixStatuses": [],
"allDivisionsApproved": false,
"allMotorsApproved": false,
"pendingMotorCount": 1,
"approvedMotorCount": 0,
"rejectedMotorCount": 0,
"totalMotorCount": 2,
"submittedBy": "EMP001",
"submittedAt": "2026-06-01T20:30:00Z"
}
}

```

## Update - Extra errors

### Form not found

```

{
  "success": false,
  "statusCode": 404,
  "message": "QC division form not found",
  "error": {
    "code": "QC_FORM_NOT_FOUND",
    "details": "No QC form exists for formId QC-F-2026-0001"
  },
  "data": null
}

```

### Unit not editable

```

{
  "success": false,
  "statusCode": 409,
  "message": "Unit already submitted for approval",
  "error": {
    "code": "UNIT_NOT_EDITABLE",
    "details": "Motor MOTOR-01 in CASTING is WAITING_FOR_APPROVAL and cannot be updated"
  },
  "data": null
}

```

## Final approval not ready

```
{
  "success": false,
  "statusCode": 422,
  "message": "Final approval not ready",
  "error": {
    "code": "FINAL_APPROVAL_NOT_READY",
    "details": {
      "message": "All divisions must be APPROVED before formSubmissionType SUBMIT",
      "pendingDivisions": [
        { "division": "CASTING", "pendingMotorIds": ["MOTOR-02"] },
        { "division": "MIXING", "pendingPremixNos": [2] }
      ]
    }
  },
  "data": null
}
```

## 3. Details Form

POST /api/v1/user/qc-division/details

### Request

UI currently sends:

```
{
  "formId": "QC-F-2026-0001",
  "subDepartmentId": 115
}
```

(subDepartmentId is required by current FE.)

### Success - multi-division form with partial statuses

- divisionDetails must match create/update (units + sections), plus status fields.
- Always include motorStatuses / premixStatuses and divisionStatuses (UI partial nav + division chips).

```
{
  "success": true,
  "statusCode": 200,
  "message": "QC division form details fetched successfully",
  "timestamp": "2026-06-01T21:00:00Z",
  "data": {
    "formId": "QC-F-2026-0001",
    "batchId": "BATCH-2026-001",
    "subDepartmentId": 115,
    "formSubmissionType": "DRAFT",
    "status": "WAITING_FOR_PARTIAL_APPROVAL",
    "allDivisionsApproved": false,
    "pendingDivisionCount": 1,
    "approvedDivisionCount": 1,
    "totalDivisionCount": 2,
    "allMotorsApproved": false,
    "pendingMotorCount": 1,
    "approvedMotorCount": 1,
    "rejectedMotorCount": 0,
    "inProgressMotorCount": 0,
    "totalMotorCount": 2,
    "divisionStatuses": [
      {
        "division": "MIXING",
        "subType": null,
        "status": "APPROVED",
        "allUnitsApproved": true,
        "approvedUnitCount": 2,
        "totalUnitCount": 2
      },
      {
        "division": "CASTING",
        "subType": null,
        "status": "WAITING_FOR_PARTIAL_APPROVAL",

```

```

    "allUnitsApproved": false,
    "pendingUnitCount": 1,
    "approvedUnitCount": 1,
    "waitingForApprovalUnitCount": 1,
    "totalUnitCount": 2
  }
],
"motorStatuses": [
  {
    "division": "CASTING",
    "motorId": "MOTOR-01",
    "motorSubmissionType": "SUBMIT",
    "motorSubmissionStatus": "APPROVED",
    "remarks": "Approved",
    "rejectionReason": null,
    "actionBy": "APR001",
    "actionAt": "2026-06-01T20:45:00Z"
  },
  {
    "division": "CASTING",
    "motorId": "MOTOR-02",
    "motorSubmissionType": "SUBMIT",
    "motorSubmissionStatus": "WAITING_FOR_APPROVAL",
    "remarks": null,
    "rejectionReason": null,
    "actionBy": null,
    "actionAt": null
  }
],
"premixStatuses": [
  {
    "division": "MIXING",
    "stageType": "PREMIX",
    "premixNo": 1,
    "premixSubmissionType": "SUBMIT",
    "premixSubmissionStatus": "APPROVED"
  },
  {
    "division": "MIXING",
    "stageType": "FINAL_MIX",
    "premixNo": 1,
    "premixSubmissionType": "SUBMIT",
    "premixSubmissionStatus": "APPROVED"
  }
],
"createdBy": "EMP001",
"createdAt": "2026-06-01T18:00:00Z",
"submittedBy": "EMP001",
"submittedAt": "2026-06-01T20:00:00Z",
"divisionDetails": [
  {
    "division": "MIXING",
    "subType": null,
    "status": "APPROVED",
    "data": {
      "premixes": [
        {
          "premixNo": 1,
          "stageType": "PREMIX",
          "premixSubmissionType": "SUBMIT",
          "premixSubmissionStatus": "APPROVED",
          "sections": [
            { "sectionId": "PREMIX_DETAILS", "sectionData": [{ "...": "..."} ] }
          ]
        }
      ],
      {
        "premixNo": 1,
        "stageType": "FINAL_MIX",
        "premixSubmissionType": "SUBMIT",
        "premixSubmissionStatus": "APPROVED",
        "sections": [
          { "sectionId": "FINAL_MIX_DETAILS", "sectionData": [{ "...": "..."} ] },
          { "sectionId": "VISCOSITY_BUILD_UP", "sectionData": [{ "...": "..."} ] }
        ]
      }
    ]
  }
],
{
  "division": "CASTING",

```

```

"subType": null,
"status": "WAITING_FOR_PARTIAL_APPROVAL",
"data": {
  "motors": [
    {
      "motorId": "MOTOR-01",
      "motorSubmissionType": "SUBMIT",
      "motorSubmissionStatus": "APPROVED",
      "remarks": "Approved",
      "rejectionReason": null,
      "actionBy": "APR001",
      "actionAt": "2026-06-01T20:45:00Z",
      "sections": [
        { "sectionId": "CASTING_SELECTION", "sectionData": [{ "...": "same as create" }] },
        { "sectionId": "FINAL_ASSEMBLY", "sectionData": [{ "...": "same as create" }] },
        { "sectionId": "PROPELLANT_CASTING", "sectionData": [{ "...": "..." }] },
        { "sectionId": "WEIGHTMENT_DETAILS", "sectionData": [{ "...": "..." }] },
        { "sectionId": "POST_CAST_OPERATIONS", "sectionData": [{ "...": "..." }] }
      ]
    },
    {
      "motorId": "MOTOR-02",
      "motorSubmissionType": "SUBMIT",
      "motorSubmissionStatus": "WAITING_FOR_APPROVAL",
      "remarks": null,
      "rejectionReason": null,
      "actionBy": null,
      "actionAt": null,
      "sections": []
    }
  ]
}
}
}
}
}

```

### After final proceed (formSubmissionType: "SUBMIT")

```

"formSubmissionType": "SUBMIT",
"status": "WAITING_FOR_COMPLETE_APPROVAL",
"allDivisionsApproved": true,
"allMotorsApproved": true

```

### After form-level approver approve

```

"status": "APPROVED"

```

(or FINAL\_APPROVAL\_COMPLETED if that is your batch terminal status - same as Case Prep).

## Details - Errors

Keep 400 / 401 / 403 / 404 / 500. Validation example:

```

{
  "success": false,
  "statusCode": 400,
  "message": "Invalid request",
  "error": {
    "code": "INVALID_REQUEST",
    "fieldErrors": [
      { "field": "formId", "message": "Form Id is required" }
    ]
  },
  "data": null
}

```

## Section ID checklist (by division)

Use schema section.id values as sectionId. Examples from current QC schema docs:

| Division                  | Typical sectionIds   |
|---------------------------|----------------------|
| RAW_MATERIAL_REVALIDATION | RAW_MATERIAL_DETAILS |

|                           |                                                                        |
|---------------------------|------------------------------------------------------------------------|
| RAW_MATERIAL_PROCESSING   | solid/liquid processing section ids from schema                        |
| MIXING + PREMIX           | PREMIX_DETAILS                                                         |
| MIXING + FINAL_MIX        | final-mix details + viscosity sections                                 |
| HARDWARE + ABRADING       | ABRADING_DETAILS                                                       |
| HARDWARE + PREHEATING     | preheating section ids                                                 |
| HARDWARE + LINEAR_COATING | liner coating section ids                                              |
| HARDWARE + DISPATCH       | dispatch section ids                                                   |
| CASTING                   | CASTING_SELECTION, FINAL_ASSEMBLY, propellant/weightment/post-cast ids |
| CURING                    | curing selection + cycle sections (per NORMAL/CONFINED/N2_PRESSURE)    |
| DE_CORING                 | de-coring section ids                                                  |
| TRIMMING                  | trimming measurement section ids                                       |
| POST_CURE                 | loose-flap / inhibition section ids                                    |
| NDT                       | radiography / visual / media section ids                               |
| PROPELLANT_PROPERTIES     | mechanical / interface / burn-rate / ballistic section ids             |
| WEIGHTMENT                | weighscale + motor weight section ids                                  |

No motorNo arrays inside sections - one motor/premix = one unit entry with its own sections[].

## Flow summary (Case Prep / RMP mirror + division)

| Action                                                     | Unit status          | Division status                    | Form status                              |
|------------------------------------------------------------|----------------------|------------------------------------|------------------------------------------|
| Create/Update unit DRAFT                                   | IN_PROGRESS          | IN_PROGRESS                        | DRAFT / IN_PROGRESS                      |
| Update unit SUBMIT                                         | WAITING_FOR_APPROVAL | WAITING_FOR_PARTIAL_APPROVAL       | WAITING_FOR_PARTIAL_APPROVAL             |
| Approver unit approve                                      | APPROVED             | stays partial until all units done | stays partial                            |
| All units in division approved                             | -                    | APPROVED                           | still partial if other divisions pending |
| All divisions approved + Update formSubmissionType: SUBMIT | -                    | all APPROVED                       | WAITING_FOR_COMPLETE_APPROVAL            |
| Approver form approve                                      | -                    | -                                  | APPROVED / FINAL_APPROVAL_COMPLETED      |

## FE <-> API notes (for backend implementers)

1. Today FE create/update still sends flat data: { sections } without motorSubmissionType. Contract above is the target; FE should add unit wrappers + \*SubmissionType (same as Case Prep motors[].motorSubmissionType).
2. Details must return unit lists under each division so mapDivisionDetailsToPartialNav can build motor/premix nav.
3. Always stamp division on every motorStatuses / premixStatuses row - same MOTOR-01 can exist under CASTING and CURING independently.
4. Create once per batch; every later unit save is update.

\*Document generated for QC Division partial-approval API contract.\*